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RINO

A Multi-Institution Team Proposes Treating All Vision Tasks as RGB-to-RGB Image Editing, Enabling One Backbone to Handle Depth, Segmentation, and Normals Without Fine-Tuning

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The dominant paradigm in computer vision maintains separate models for separate tasks: one model for depth, another for segmentation, a third for surface normals. Even modern unified models typically require task-specific heads or adapters and must be fine-tuned per benchmark. RINO (RGB In, RGB Out) proposes a different architecture: encode every visual output as an RGB image and reduce every vision task to an RGB-to-RGB image editing problem, sharing all parameters across all tasks without task-specific components.

AT A GLANCE

Problem: Dense visual perception requires separate models per task; existing unified approaches still use task-specific heads or require benchmark fine-tuning.

Why it matters: True generalist vision requires a single set of parameters that transfers across tasks through a unified interface, as text does for NLP.

Method: All visual modalities (depth, masks, normals, keypoints) encoded as RGB images; tasks become text-conditioned RGB-to-RGB editing problems.

Result: Competitive zero-shot performance on four dense perception benchmarks using a single generic image editing backbone.

Evidence: NYU-Depth-v2, ADE20K segmentation, ScanNet normals, H3WB keypoints; zero-shot, no benchmark fine-tuning.

The Big Picture

Language models achieve generality through a single universal representation: text. Any NLP task — translation, summarization, QA, classification — can be expressed as a text-in, text-out problem, allowing one set of parameters to transfer across all of them. RINO asks whether pixels can serve the same function for vision.

The central observation is that all dense vision outputs are already spatial arrays over the same 2D grid as the input image. Depth is a scalar per pixel, surface normals are a 3-vector per pixel, segmentation masks are a class label per pixel, and keypoint heatmaps are a confidence per pixel. All of these can be rendered as an RGB image and decoded back without loss, if the encoding is chosen carefully.

Why Existing Approaches Fall Short

Task-specific models are powerful but expensive: each benchmark requires a separate model, separate fine-tuning run, and separate inference process. Multi-task models (e.g., UniPerceiver, InternImage) unify multiple tasks in one architecture but still require task-specific output heads that must be trained on benchmark data. Generalist models like Painter and PromptDiffusion reduce fine-tuning requirements but retain task-specific structured output formats.

The key gap is that no prior work treats the task output itself as RGB data fed into a backbone that was not designed for the task at all. RINO demonstrates that a backbone trained entirely on natural image editing — with no exposure to depth maps, segmentation masks, or surface normals — can perform these tasks zero-shot if the output is presented to it as an RGB image to edit.

Core Mechanism

Each visual modality is assigned a lossless or near-lossless RGB encoding:

1. **Depth maps:** Depth values d are normalized to $[0, 255]$ and stored in the red channel. The green channel encodes epistemic uncertainty; blue is zero.
2. **Segmentation masks:** Each semantic class receives a fixed, perceptually distinct RGB color. Instance masks use hue-shifted variants of the class color, enabling visual discrimination of instances.
3. **Surface normals:** The XYZ components of the unit normal vector are linearly mapped to $[0, 255]$ triplets: $\text{RGB} = \lfloor 127.5(\mathbf{n} + \mathbf{1}) \rfloor$.
4. **Keypoint heatmaps:** Per-joint Gaussian confidence heatmaps are stacked into the RGB channels (up to three joints per RGB image, with multiple images for larger joint sets).

A text instruction specifies the desired output type. The editing backbone processes the input image and instruction, producing an RGB output that is decoded into the task-specific representation.

Technical Formulation

For a task T with input image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$ and ground truth output $\mathbf{y}_T$, RINO defines an encoding function $\phi_T : \mathbf{y}_T \rightarrow \mathbb{R}^{H \times W \times 3}$ and decoding function ϕ_T^{-1} . The backbone f_θ is trained to minimize:

$$\mathcal{L} = \mathbb{E}_{(x, y_T, T)} \|f_\theta(\mathbf{x}, \tau_T) - \phi_T(\mathbf{y}_T)\|_2^2$$

where τ_T is the text instruction for task T . Crucially, f_θ is the pretrained image editing backbone — its weights are frozen for zero-shot evaluation.

Experimental Findings

Task	Metric	RINO	Fine-Tuned
Depth (NYU-v2)	AbsRel↓	0.137	0.103
Seg. (ADE20K)	mIoU↑	38.7	47.2
Normals (ScanNet)	Mean Ang.↓	11.3°	9.1°
Keypoints (H3WB)	PCK↑	61.4	68.9

RINO’s zero-shot numbers are within 15–25% (relative) of dedicated fine-tuned models on each task, using a single set of backbone weights that was never trained on any of these datasets.

Ablations and Interpretation

Task-specific heads: Adding a lightweight task-specific head (while keeping the backbone frozen) closes approximately half the gap to fully fine-tuned models on depth and segmentation. This confirms that the backbone captures the necessary spatial representations; the remaining gap is in task-specific prediction.

RGB encoding quality: Using a uniform grayscale depth encoding (instead of the normalized red-channel scheme) degrades depth AbsRel by 0.021 absolute, confirming that encoding choices matter substantially.

Instruction specificity: Vague instructions (“transform this image”) yield near-random outputs; specific instructions (“generate a surface normal map with outward normals colored red-green-blue”) approach zero-shot quality, suggesting the backbone uses the instruction as a strong conditioning signal.

Joint training benefit: When the backbone is fine-tuned jointly across depth + normals + segmentation (vs. separately), each task improves by 1–2%, suggesting mutual regularization from shared RGB representations.

Reference

Timing Yang et al. **Let RGB Be the Language of Vision**. arXiv:2607.12450, July 2026. <https://arxiv.org/abs/2607.12450>